

Amazon CloudWatch - Service Overview

Introduction

Amazon CloudWatch is a monitoring and observability service that collects and tracks metrics, collects and monitors log files, sets alarms, and automatically reacts to changes in AWS resources. CloudWatch provides a unified view of operational health across AWS resources, applications, and services.

Metrics

A metric is a time-ordered set of data points representing some measurable quantity, such as CPU utilization, network throughput, or a custom application-level value like queries processed per second. AWS services automatically publish metrics to CloudWatch at various resolutions, typically every 1 or 5 minutes, with detailed monitoring providing 1-minute granularity for an additional cost on some services.

Custom Metrics

Applications can publish their own custom metrics to CloudWatch via the PutMetricData API, enabling monitoring of business or application-specific values not natively tracked by AWS, such as retrieval latency in a RAG pipeline, tokens consumed per query, or the number of documents ingested.

CloudWatch Logs

CloudWatch Logs allows you to centralize logs from EC2 instances, Lambda functions, ECS tasks, and other sources into a single, searchable location. Logs are organized into log groups and log streams. The `awslogs`` driver used by ECS tasks, for example, sends container stdout/stderr directly to a specified CloudWatch Logs log group with no additional agent required.

Logs Insights

CloudWatch Logs Insights provides a purpose-built query language for interactively searching and analyzing log data, allowing structured queries against JSON-formatted log entries, such as filtering all log lines where a "level" field equals "ERROR" within a specific time range.

Alarms

A CloudWatch Alarm watches a single metric (or a math expression combining metrics) over a specified time period, and performs an action when the metric breaches a defined threshold. Actions can include sending a notification via SNS, triggering an Auto Scaling action, or invoking a Lambda

function for automated remediation.

Dashboards

CloudWatch Dashboards allow you to create customizable home pages in the CloudWatch console that display graphs and metrics for the resources you care about, in a single view, without needing a separate third-party monitoring tool for basic operational visibility.

CloudWatch Events / EventBridge

Originally called CloudWatch Events, this functionality (now largely superseded by Amazon EventBridge) allows you to respond to state changes in AWS resources, or on a fixed schedule, by routing the event to a target such as a Lambda function or an SNS topic.

Retention

Log group retention is configurable, ranging from 1 day to indefinite retention (never expire). By default, newly created log groups retain logs indefinitely unless a retention policy is explicitly set, which can lead to unexpectedly high storage costs if not managed.

Embedded Metric Format

Rather than making a separate PutMetricData API call for every custom metric (which adds latency and API cost to the request path), applications can emit a specially structured JSON log line following CloudWatch's Embedded Metric Format (EMF). CloudWatch Logs automatically extracts metrics from EMF-formatted log entries in the background, letting an application log a single structured line containing both a human-readable event and a machine-extractable metric value (for example, `retrieval_latency_ms`) without any extra network call.

Composite Alarms

A composite alarm combines the state of multiple underlying alarms using AND/OR logic, reducing alarm noise. For example, a composite alarm might only page an on-call engineer if BOTH error rate is elevated AND request volume is above a minimum threshold, avoiding false pages during low-traffic periods when a single failed request would otherwise spike the error rate percentage.

Common Use Cases

- Centralized application logging for containerized microservices
- Alerting on-call engineers when error rates or latency exceed acceptable thresholds
- Auto Scaling Group scaling decisions driven by CPU or custom metric thresholds
- Auditing and debugging production incidents via Logs Insights queries

Best Practices

- Emit structured (JSON) log lines from applications so CloudWatch Logs Insights queries can filter on specific fields rather than parsing free text.
- Set explicit log retention periods on all log groups to control long-term storage costs.
- Define alarms for the metrics that actually predict user-facing problems (e.g. p99 latency, error rate), not just infrastructure-level metrics like raw CPU.